
INSTANT
MUSICAL
SUPER
POWERS

BESTIVAL 2026

The Royal Society Summer Science Exhibits on Tour

Program developed by

Edward Cambell
Zenon Olenski

Featuring contributions from

David Shervill
Philip Watson
Ross Elliott
Chris Olenska
Liz McLeman

In association with SOUND CONNECTIONS

In association with, And with the support of

Sound Connections

AMIC - Accessible Musical Instrument Collection

RIX
Inclusive Research Institute

DMLabs

The Royal Society

MASSIG - MIDI Association Special Standards Interest Group

Index

Index 4
Preface 6
Purpose 7
This Guide 8
Desired Outcomes 9
Key Skills 10
Key Takeaways 11
Foundations 12
Group sizes 13
FOUNDATIONS
Foundations 15
Operation Guide 16
How to create sounds 17
How to choose notes 18
Selecting low notes 19
Selecting high notes 20
Selecting notes 21
Select low octaves 22
Select higher octaves 23
Selecting octave 24
Selecting white keys 25
Selecting black keys 26
Selecting incidentals 27
Movement Guide 28
Multiplayer Guide 29
LESSON PLAN
Lesson Plan Ideas 31
Preflight Checklist 32
Suggested Schedule 33
Topics to discuss 35
Topics to explore 36
Engagement Ideas 43
What face do you make when... 45
Continue at home 52
Plenary 53
Emotional Glossary 54

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Happiness Glossary 55
Sadness Glossary 56
Despair Glossary 57
Final Checklist 58
Feedback
Initial Student Moods 61
Review 62
Worksheet 64
Student Questions 67
Student Feedback 68
Other Feedback 69
Student Feelings 71
Operation
Play Modes 73
Chromatic Mode 74
Sympathetic Mode 75
Pan Audio 76
Circle of Fifths Mode 77
Equipment
Setup 79
Tech Requirements 80
Setup Considerations 81
Screen / Projector 82
Sound 84
Electronic Equipment 86
Camera Positioning 90
Support 102
Support Directory 103
Music Communities 104
Social Enterprises 106
Music Hubs 108
Music Charities 109
Specialist Schools 113
Music Workshops 115
Copyright 116
References 117
Support 118
PhotoSYNTH 118
Smile Powered Music 118

Preface

This lesson plan aims to create a session that fosters creativity, teamwork, and an appreciation for music through immersive technologies, making learning accessible and enjoyable for students with all types of educational needs.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Purpose

The PhotoSYNTH is a fun multiplayer collaborative musical instrument that encourages group play, social interactions and cohesion.

It can be used by up to four people at the same time who use their facial expressions to make music together. With a “magic mirror” interface, people see their familiar selves which is both reassuring and disarming, encouraging playful behaviour, confident experimentation, and expressive collaboration.

With optional manned cooperative control, an overseer chooses instruments, control modes, tempo and visuals on a per-person basis, then actively guides the music through instructions and fun exercises.

This Guide

The following plans should be seen as suggestions for the foundations of a lesson with additional materials to take home to continue the practise.

It has been designed to cover the foundations and requirements, but is broad and flexible enough to form the skeleton for other sessions.

All steps and activities should be tailored for the requirements and needs of the group and individuals.

Each activity can be adapted for differing lengths of time and numbers of students.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Desired Outcomes

We believe that everybody enjoys making music, they just might not have realised it yet.

This session is part educational and part expression. We provide the tools and facilities to experiment and make music in an easy, fun way.

Through making music we teach and discover together the joy of collaboration and expression.

Enhance emotional, cognitive, and social well-being by providing an independent, immersive, intuitive and fun way for self and group expression.

What starts as a love for expression through art, can lead to a love for music theory and practise.

Key Skills

- Collaboration
- Coordination
- Confidence
- Communication
- Cognitive engagement
- Social cohesion
- Independence
- Regulation
- Life skills
- Rhythm
- Self expression
- Therapeutic expression
- Foundation in music theory

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Key Takeaways

- Joy
- Community
- Enrichment
- Cognitive engagement
- Self expression
- Access to art
- Mood regulation
- Appreciation for music making
- Personal growth
- Cultivate music as a hobby
- Free musical instruments
- Potential lead to a career in music
- Tools & exercises for home use

Foundations

Musical expression requires three solid foundations to flourish:

1. ACCESS

Facilities, instruments, tools and techniques in a safe, welcoming and comfortable space.

2. INSPIRATION

Stimulating, Independent expression. Exploration. Experimentation. Creativity and mimicry.

3. SUPPORT

Encouragement. Introduction to new forms of expression. Supervision. Education. Validation. Creating legacies. Access to tools at home.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Group sizes

The session can facilitate groups of all sizes but is optimised for groups of four or fewer.

Classes can be split into groups, with one potential delineation being those who enjoy collaborating, and those who are less used to collaboration. One of the benefits of this technology is the social aspect that encourages group play and working together.

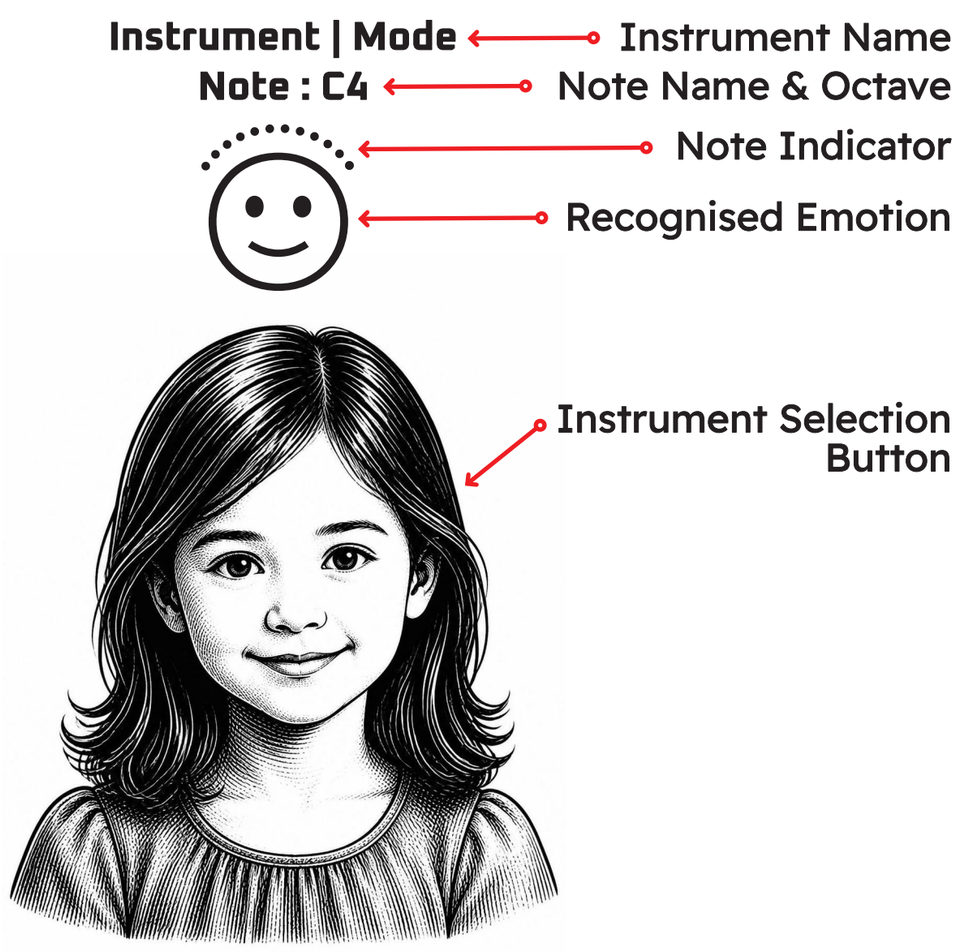
If required, a one-to-one session can be arranged with a view to building towards collaboration with a peer.

Larger sessions can revolve around a queue, handling four at a time. It is possible to have more than four players at one time, though the dynamic is very different and more competitive.

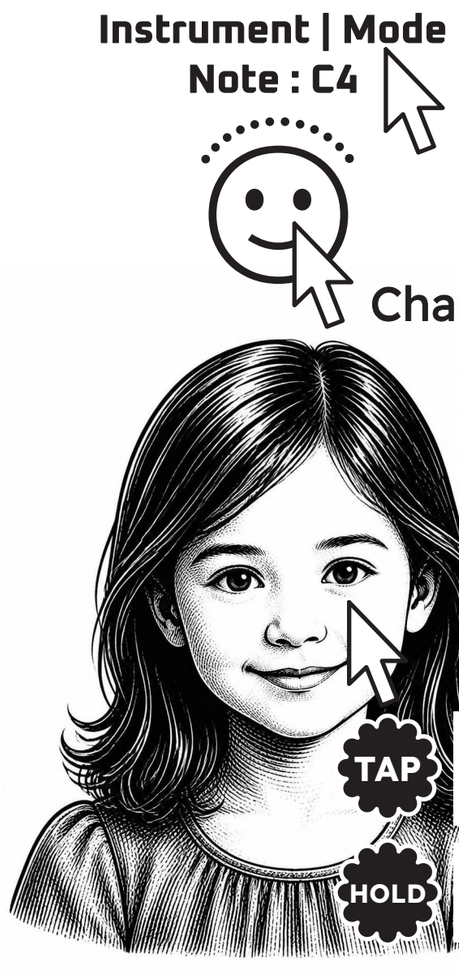
FOUNDATIONS

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Foundations
Onscreen information



Operation Guide
Controls per user



Change User Type
- Lead / Harmonic / Bass / Percussion

Change Operating Modes
- Chromatic Chords
- Circle of Fifths Chords
- Sympathetic Mode
- Circle of Fifths Arpeggio
- Video Game Score

Change Instrument
To a random one with a similar sound

Choose Instrument
From a list of all available instruments

NB. You can also swipe left or right

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

How to create sounds
Make facial expressions

Instrument Name
C4



The emoji above the user changes to represent their expression and mood.



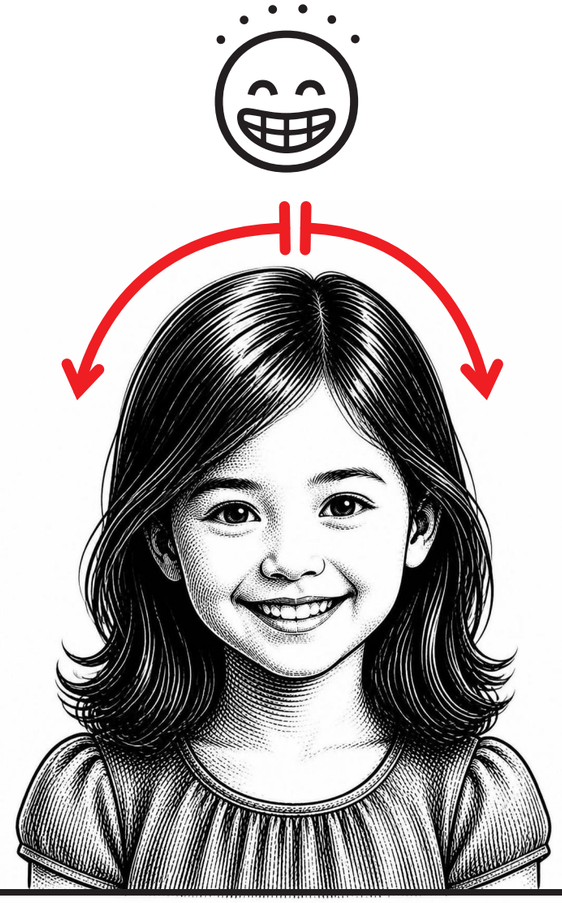
There are many different faces to unlock, each has a distinct sound and feeling.

Happy faces sound very different to frowns.

Have fun experimenting and discovering new ones!

How to choose notes
Rock head left and right

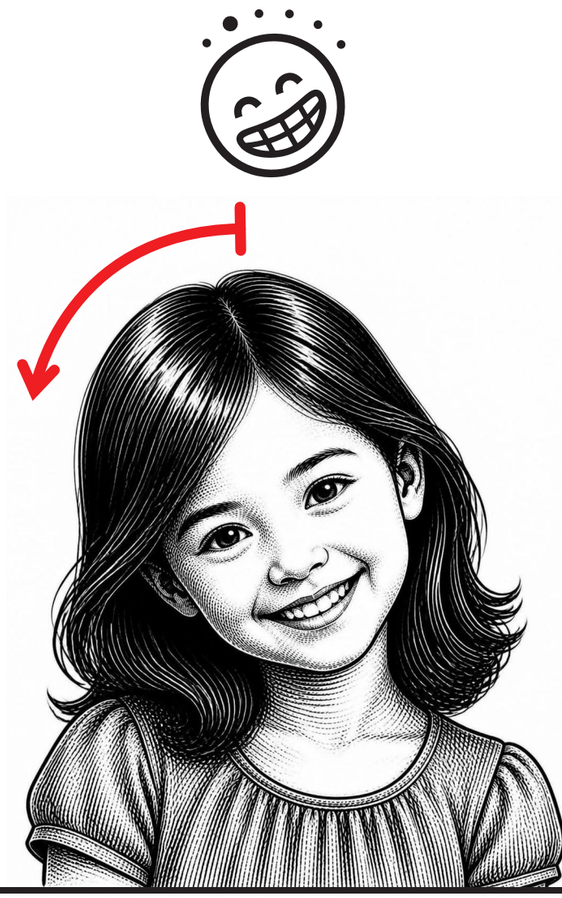
Instrument
C4



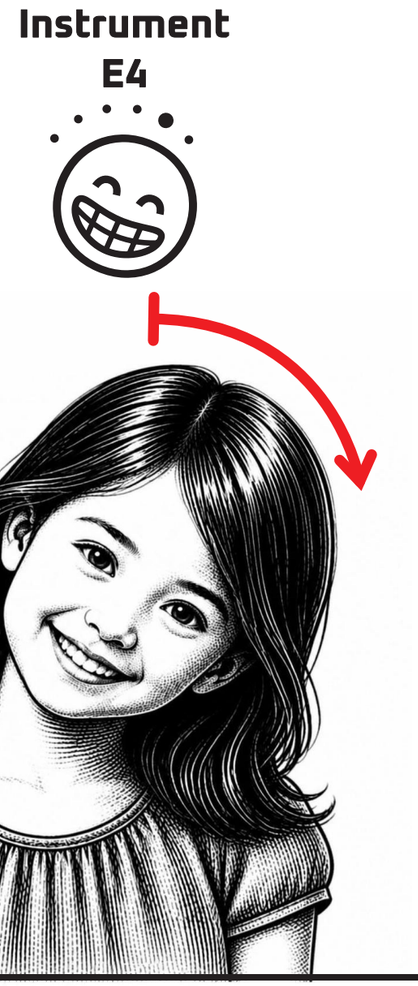
THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Selecting low notes
Crane head to the left

Instrument
B4



Selecting high notes
Crane head to the right



THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Selecting notes
Roll face side to side





Low Notes

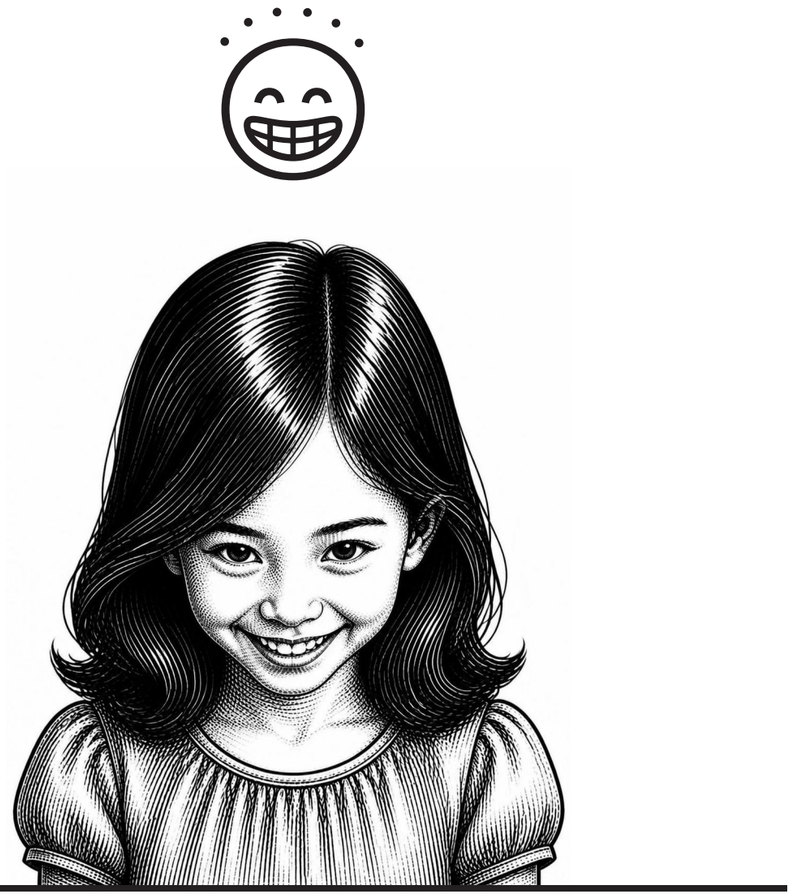


High Notes

NB. In Circle of Fifth's mode the notes play in a different sequence

Select low octaves
Point nose down

Instrument
C1



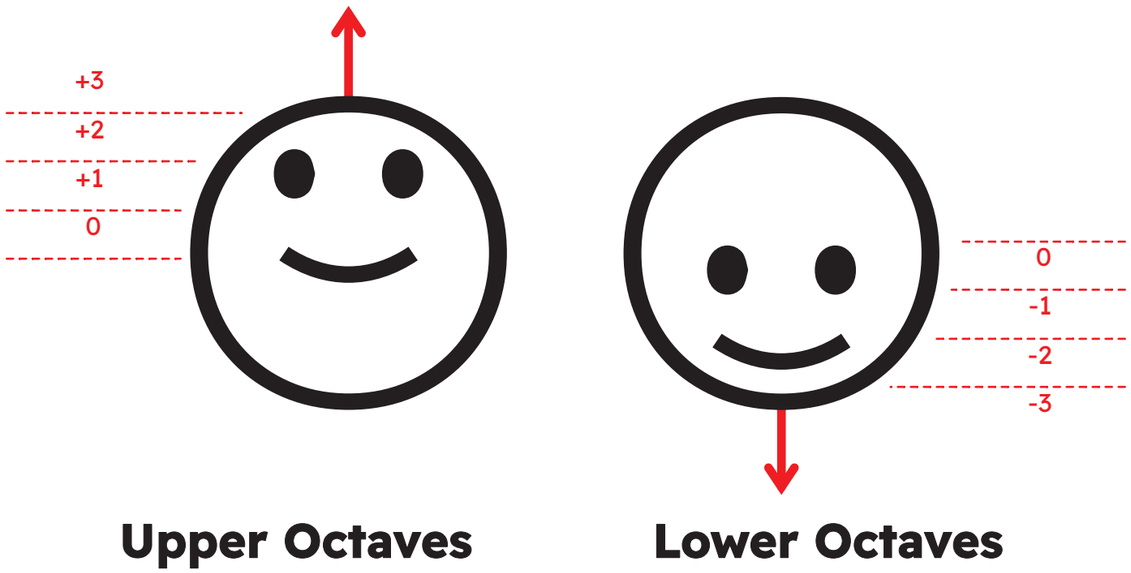
THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Select higher octaves
Point nose up

Instrument
C6

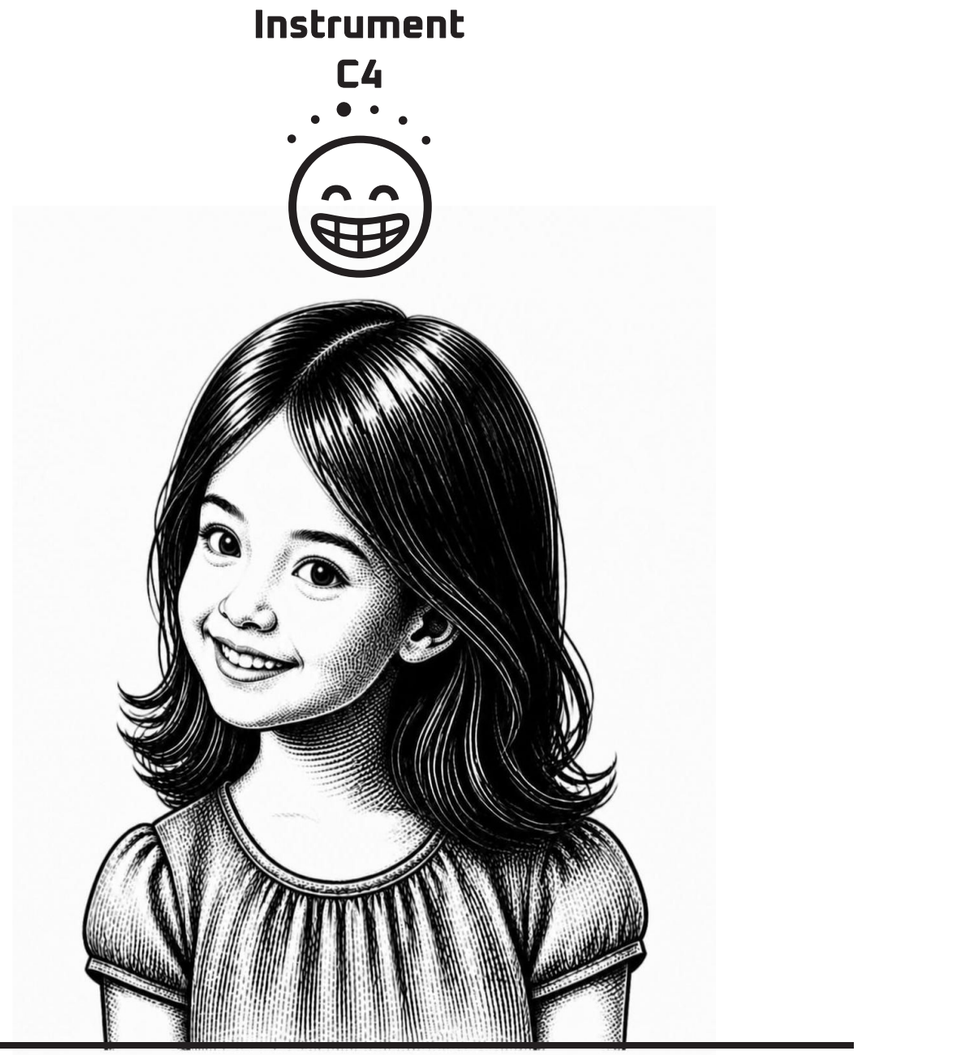


Selecting octave
Point nose up & down

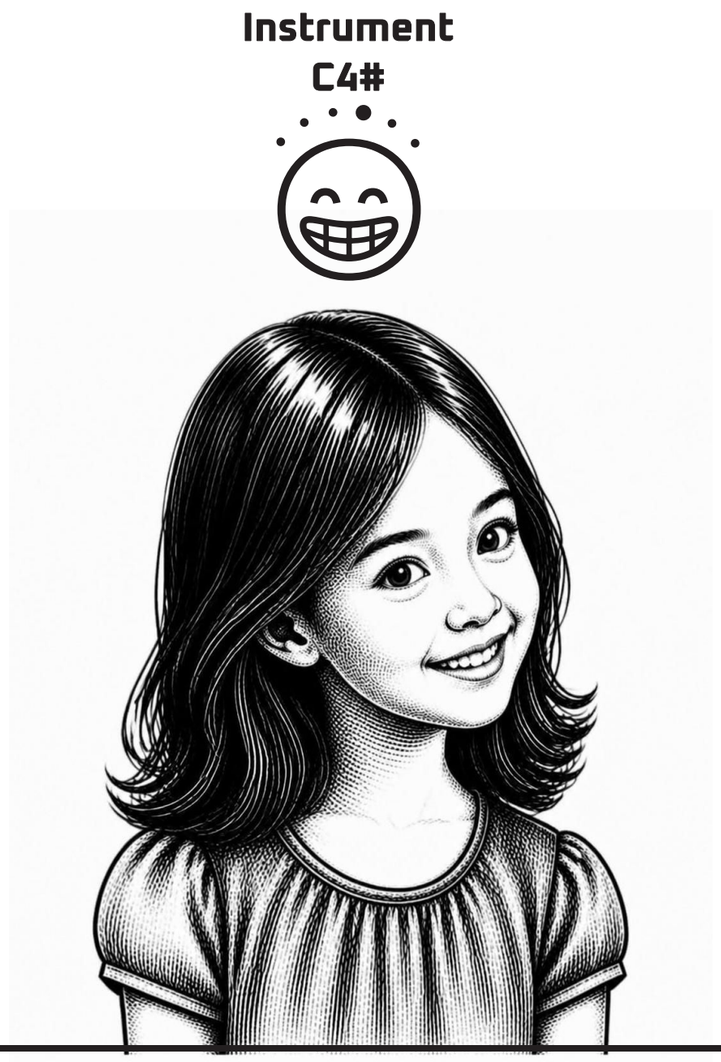


NB. Each player has access to a different octave range

Selecting white keys
Point nose right

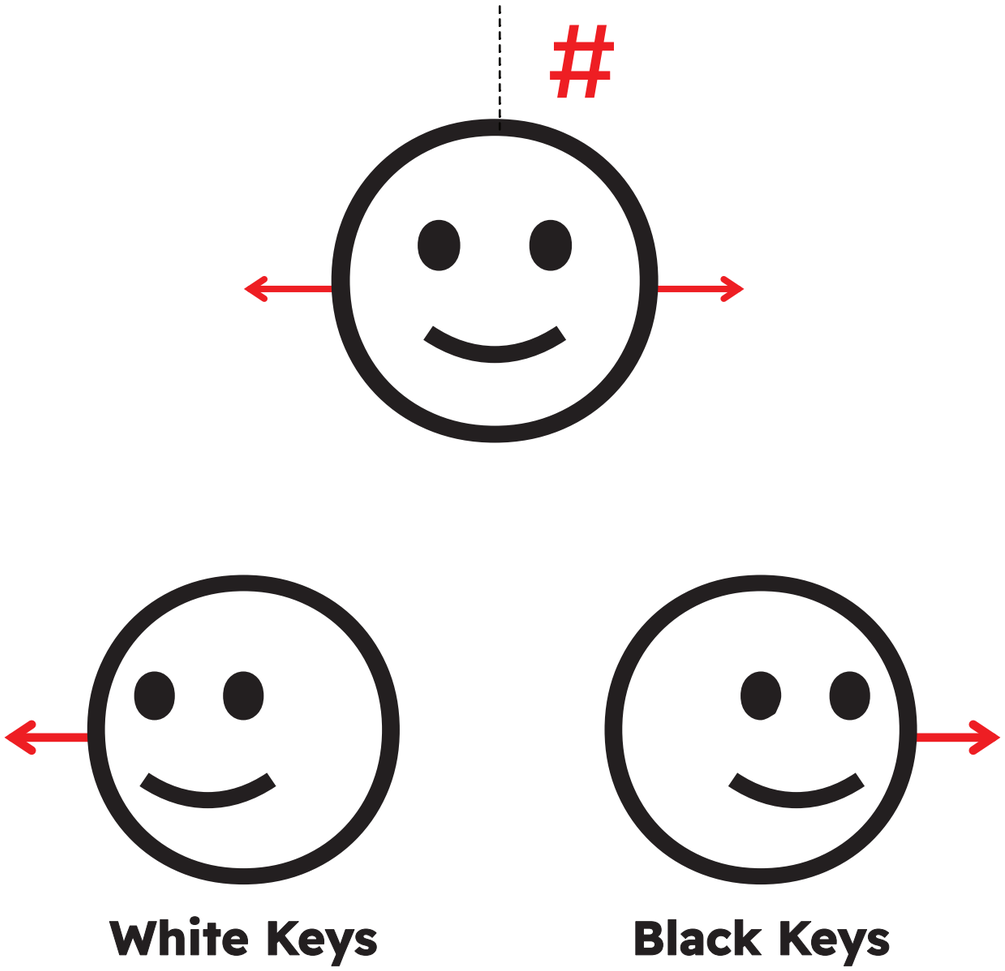


Selecting black keys
Point nose left

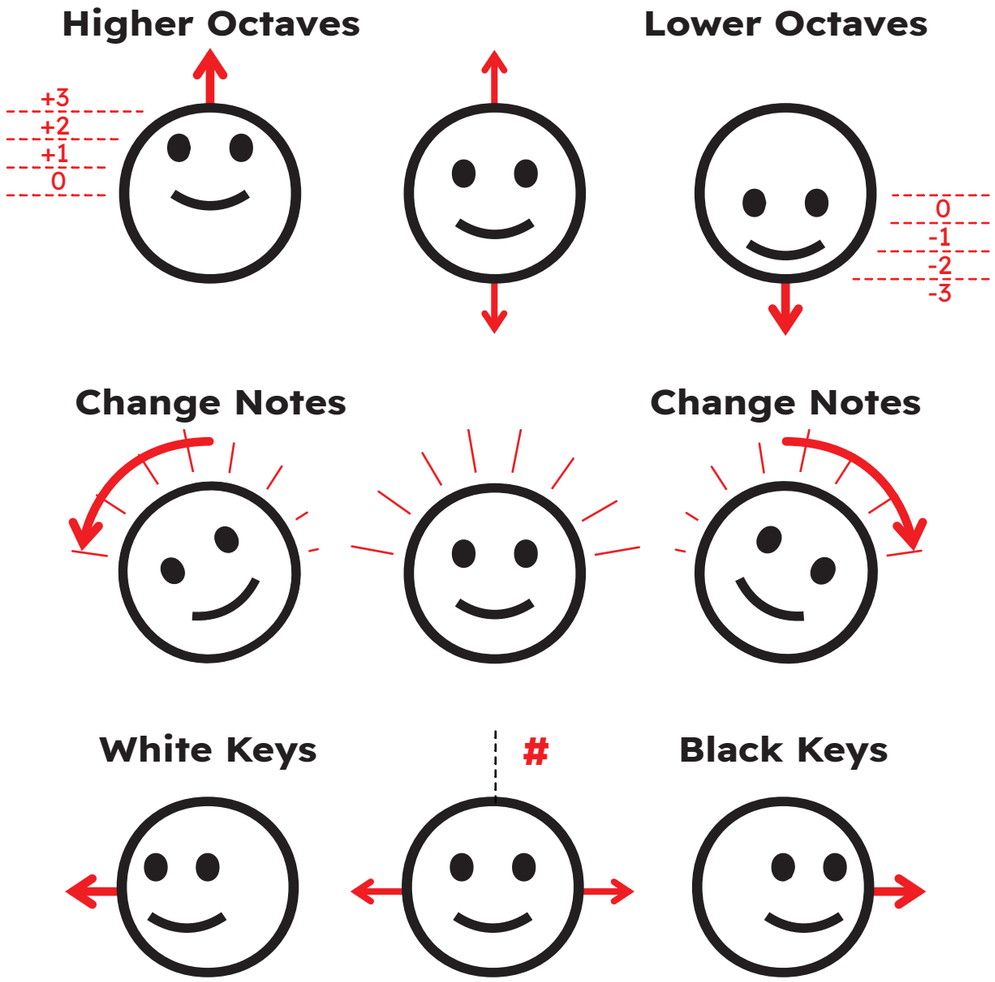


THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

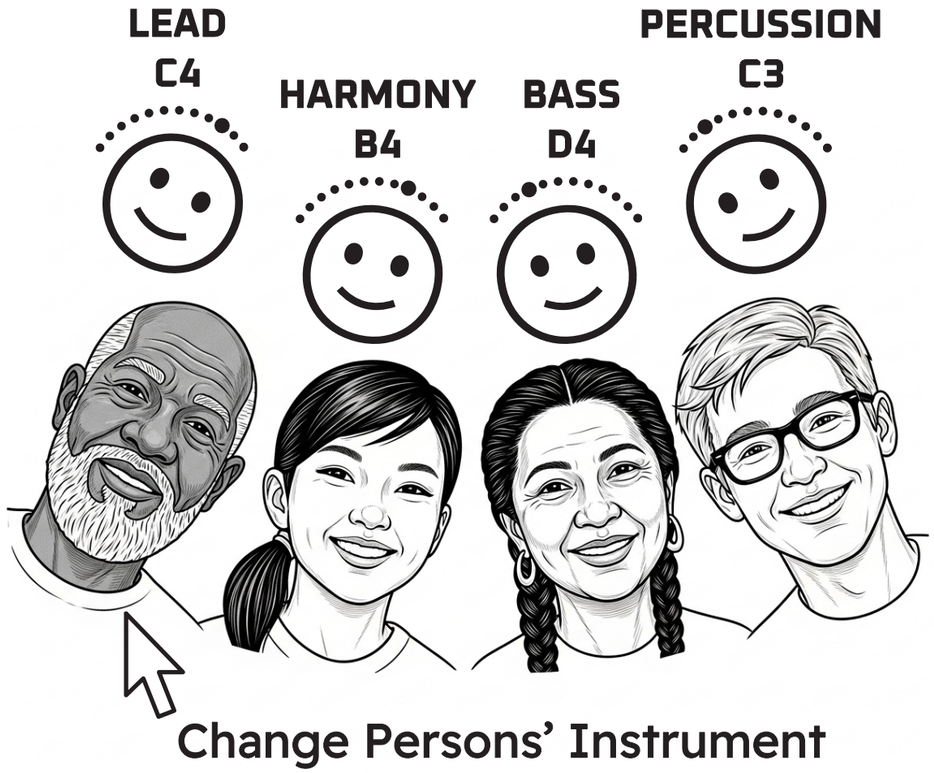
Selecting incidentals
Point nose left & right



Movement Guide



Multiplayer Guide
Roles in the band



NB. Player roles can be changed by pressing the person's buttons

LESSON PLAN

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Lesson Plan Ideas

This lesson plan aims to create a session that fosters creativity, teamwork, and an appreciation for music through immersive technologies that grant instant musical superpowers and making learning accessible and enjoyable for students with all types of requirements.

Approximate duration : 40 minutes

Age range : 4 years+

☑ Preflight Checklist

- [ ] **Ensure setup and area is safe**
  No trailing cables, nothing unnecessary
- [ ] **Sterilise all equipment**
  Wipe down with antiseptic wipes
- [ ] **Turn all equipment on & connect**
  Set volumes / scales appropriately
- [ ] **Test everything is working**
  Before the session, away from the kids
- [ ] **Ensure lighting is comprehensive**
  Solid, consistent room illumination
- [ ] **Determine group requirements**
  Modify plan & setup accordingly
- [ ] **Eradicate any and all distractions**
  Phone off, ensure clear area

Suggested Schedule



1. **ACTIVITY** (~5-10 min)
Faces and sounds (explore)
Warm up and introduction with some fun exercises and activities designed to prepare and engage.



2. **ACTIVITY** (~5-10 min)
Reflective discussion
Musical experimentation and exploration with faces and expressions with leading questioning.



3. **BRAIN BREAK** (~5 min)
Brain break
Resting, conversation and hands on play with the other instruments.



4. **MAIN SESSION** (~20 min)
Jam and collab
Create a song :
Enrichment / therapeutic expression instruments to explore.

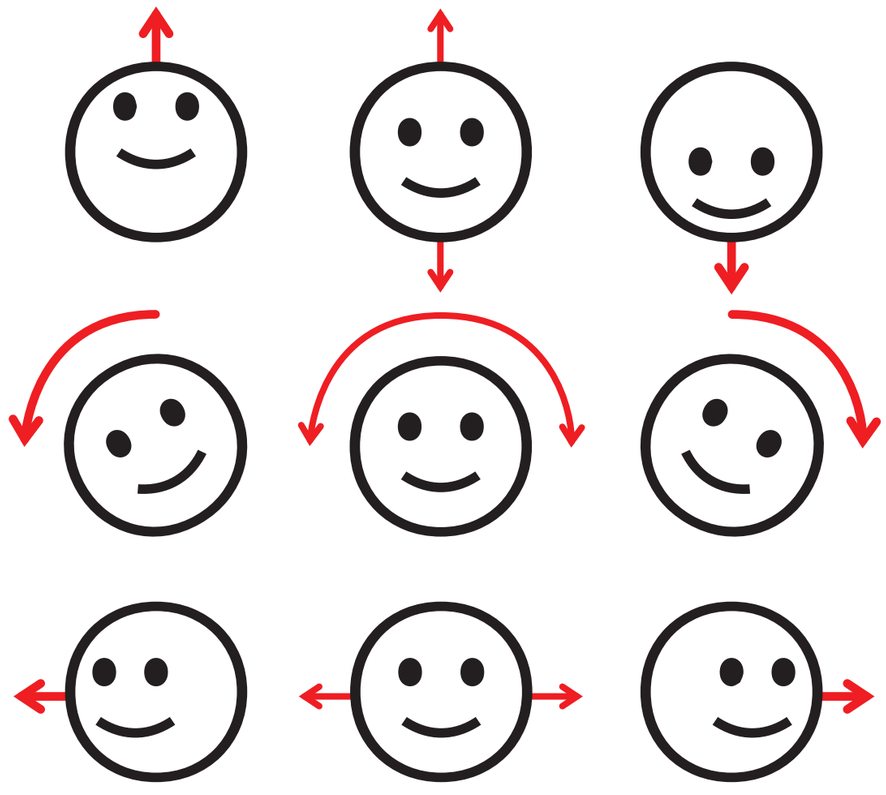


5. **PLENARY & TAKEAWAYS**
Feedback and suggestions
Fill out forms. share give-aways, data sheets and software QR codes.

1 | FACES & SOUND

Exploration
5-10 minutes

Encouraging warm-up exercises such as moving, tilting and shaking the head as well as pulling different facial expressions so as to prime the students for how movements become sounds.



THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

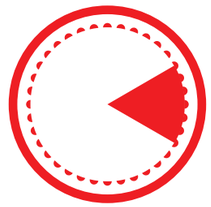
Topics to discuss

- What sounds can you make?
- What is your favourite sound?
- How do those sounds make you feel?
- What songs do you like to sing along to?
- How does singing make you feel?
- When you listen to your favourite music, how does it make you feel?
- What facial expressions can you make?
- Do you have a favourite expression?
- What kind of sounds would you like to make?
- Do you know how to play a musical instrument?

Topics to explore

- Music can express feelings.
- Listening to music affects your emotions.
- Emotions can influence how we make music.
- Making music is a good way to express yourself.
- Music making is a nice way to spend time with friends.
- Music is a language.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

2 | 

REFLECTIVE DISCUSSION

Individual expression
5-10 minutes

There's no "wrong" way to make music, every expression creates something unique!

NB. There are a range of different facial expressions over the next few pages that can be used to help with communication, as well as an emotional glossary later on.



How do these faces sound?



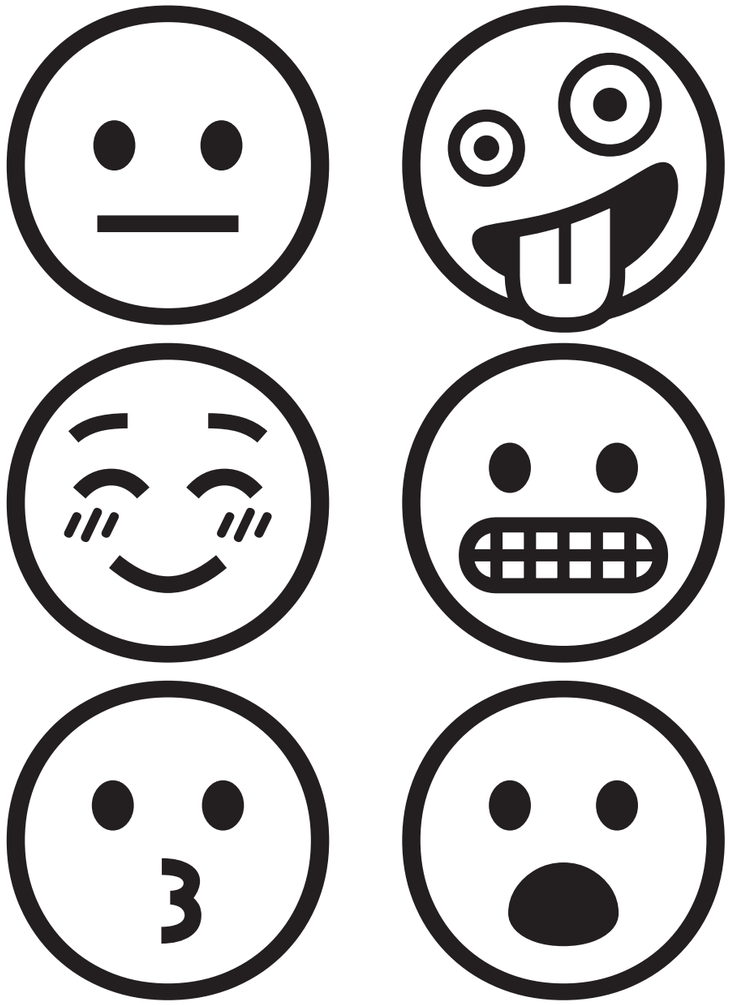
THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS



How do these faces sound?



How about these?



3 | BRAIN BREAK

Relax
~5 minutes

Rest time.

This is also an opportunity to get hands on with the other adaptive musical tech designed specifically for the Disabled community. These other instruments are all very fun, simple to use and available commercially.

You can learn more about all available technology in the reference section at the end of this manual where there are links to learn more about them.

4

JAM TIME

Experimentation
~20 minutes

Main session – Create a melody.

individually at first, can be split off to create or have 1-20 create an rotate while others engage and explore with the other accessible musical instruments.

Some possible exercises for the group :

- In pairs, students take turns making different faces while their partner observes what sounds they create.
- Individuals create a "face symphony", a sequence of 3 different expressions that tell a mini musical story.
- Groups of four jam together freely and each has a role in the band.d

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Engagement Ideas

- Take a deep breath… and relax.
- Who can make a happy face?
- Give me your biggest grin — show your teeth!
- Is that your happiest face? What might a happy sound sound like?
- Have you got a silly face?
- Make the silliest face you can! You can make one sillier than that!
- How about your sad face?
- Look up like you're trying to remember something important!
- Look like you don't believe a word I'm saying… as suspicious as possible!

- Try keep a straight face… as serious as possible… no laughing allowed!
- Shake your head like someone just said something ridiculous!
- Can you raise both eyebrows like you’ve just heard terrific news?
- Look up, is that a bird or a tiny dragon flying past?
- Squint like you’re trying to see something tiny and far away.
- Cross your eyes and stick your tongue out… Can you be sillier!?
- Pretend you are eating something very sour like a lemon!
- Put on your superhero face!

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

What face do you make when...

- ...someone gives you a big cuddle
- ...you drop your ice cream
- ...your friend makes you laugh
- ...you are waiting and waiting for your turn
- ...someone takes your toy without asking
- ...you find your favourite snack
- ...it starts raining when you want to play outside
- ...you hear a very funny noise
- ...you lose your teddy

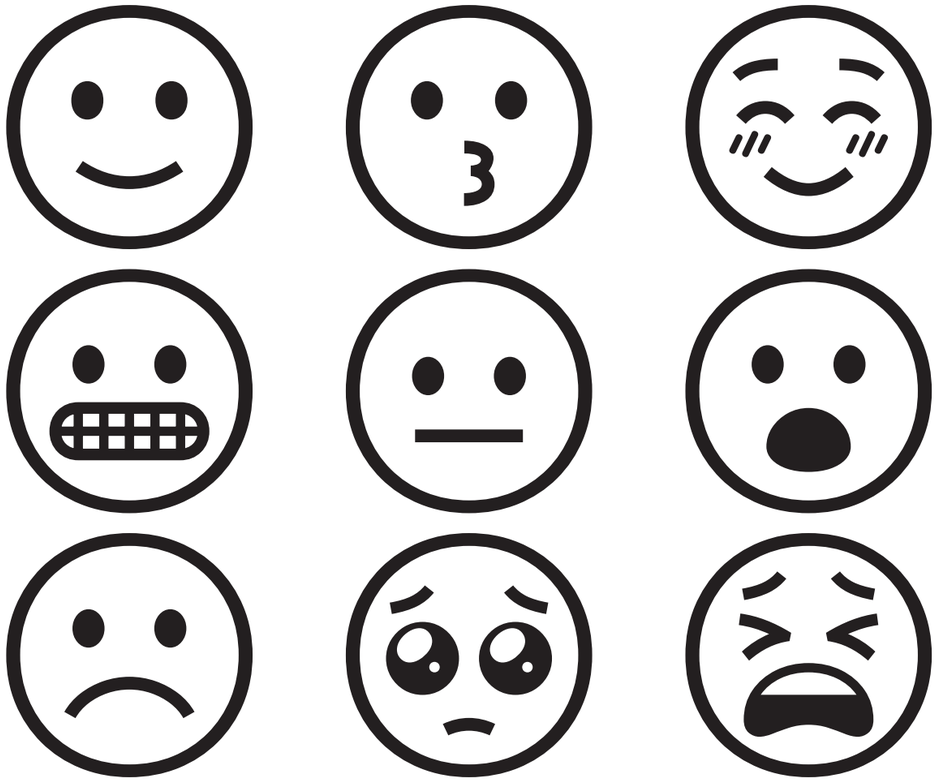
- ...you get a sticker
- ...you feel really sleepy
- ...you are trying not to laugh
- ...someone says surprise
- ...you see a dog bigger than you expected
- ...you can't find your shoe
- ...you get to jump in a puddle
- ...someone says it's time to tidy up
- ...you build a very tall tower
- ...the tower falls down
- ...you smell something yummy
- ...you taste something yucky
- ...someone shares with you
- ...someone won't share

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

- ...you hear your favourite song
- ...you are pretending to be a grumpy bear
- ...you are sleepy
- ...you are surprised

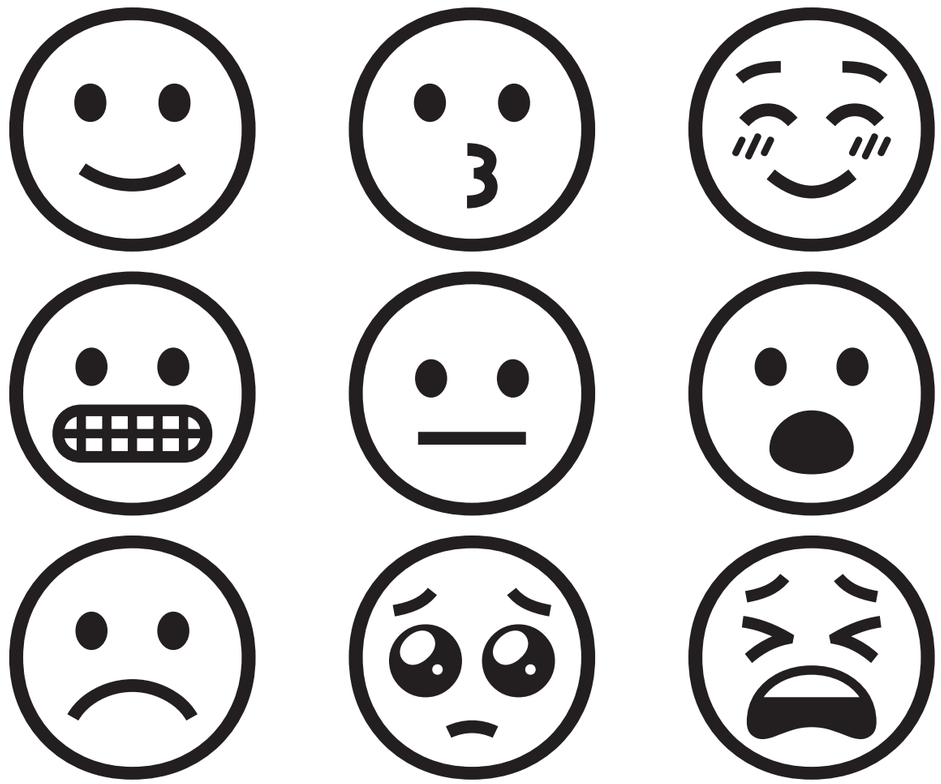
- ...a monkey steals your banana!
- ...your socks feel all wet
- ...you see the biggest cake in the world!
- ...a feather tickles your nose
- ...you are a king or queen on a throne
- ...you are a dragon trying not to sneeze
- ...you are a lion with a sore paw

What face would you make if you dropped your ice cream?

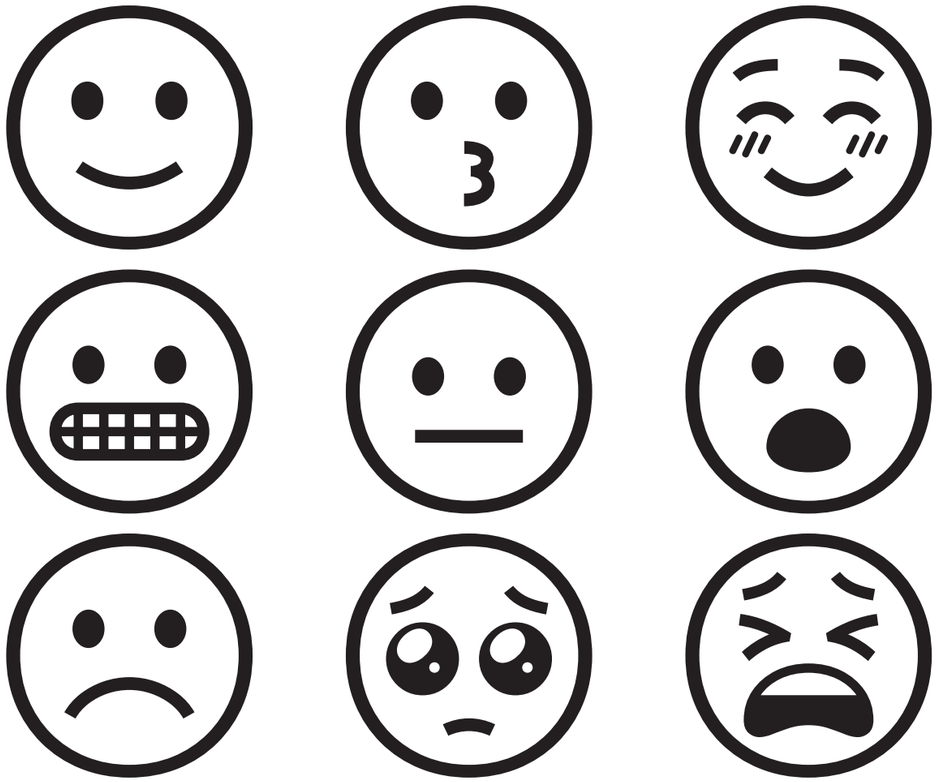


THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

When it’s bedtime and you don’t want to go to bed what face do you make?



When you feel proud because you did it by yourself what face do you make?



5 | FEEDBACK SESSION

Conversation
~10 minutes

It is essential to record progress and feedback as well as any suggestions, problems, reflections and situations you may have encountered. Successes and failures will allow us to improve and tailor this pilot into something special and unique, driven by real stories and situations.

There are some feedback forms for both practitioners / parents as well as for the students. These will allow us to understand whether we have achieved our objectives and how we can improve.

Please be as critical as possible!

Continue at home

If you would like to continue the session at home and learn more about the project, visit the following links :

Continue making music at home
Works best with Chrome or Edge browser on a modern device.
https://photosynth.app

Learn more about the project
The science, technology and music theory is explained.
https://photosynth.lol

Plenary

Once the worksheet has been completed, summarise how you feel the session went and assess the progress and development of the students, and any useful observations.

Emotional Glossary

Every person expresses themselves differently but there is a common universal language that affects all people - smiling is considered a sign of happiness for example.

The following reference tables contain concise summaries of the main expressions and moods you will encounter in these sessions.

A variety of different expressions are rewarded in the software and progress is recorded per person, allowing the subject to be monitored and their development analysed.

Happiness Glossary

😐	**NEUTRAL** Expressionless, indifferent, calm
😉	**WINK** Playful, joking, friendly
🙂	**SMILING SLIGHTLY** Gentle smile, subtle happiness
😊	**FULL SMILE** Happiness, warmth, content
☺️	**HAPPY SMILE** Contentedness, friendly, cheery
😀	**BROAD SMILE** Happy, cheerful, friendly
😃	**WIDE-EYED SMILE** Very happy, excited, joyful
😄	**ECSTATIC** Happy, delighted
😁	**GRINNING** Beaming, excited, thrilled
😆	**CHUCKLING** Laughing, amused, joyful
🤗	**HUGGING** Comfort, support, embracing
🤭	**GIGGLING** Amused, giggling, playful
🥲	**HAPPY TEARS** Bittersweet ambivalence
😌	**RELIEVED** Peaceful, relaxed, satisfied
😋	**SAVOURING** Enjoying, pleased, satisfied,
😛	**TONGUE** Playful, silly, cheeky, teasing
😝	**RASPBERRY** Goofy, joking, funny
😜	**TONGUE WINK** Playful, fun, mischievous
🥹	**THANKFUL** Emotional, moved, appreciative
😗	**KISS** Affectionate, lovely, gentle
😙	**KISS** Loving, passionate, affection
😚	**EXPECTING KISS** Cheerful kiss, happy affection
😘	**KISSING WINK** Affectionate, teasing
😍	**HEART EYES** Love, infatuation, admiration
🤩	**STAR STRUCK** Amazed, impressed, awestruck
🤪	**ZANY** Crazy, wild, absurd, silly
😇	**HALO** Angelic, innocent, well-behaved
😏	**SMIRK** Confident, steadfast, sarcastic
😮	**ASTONISHED** Shocked, amazed, beguiled
😑	**SILENT** Quiet, unimpressed, silent

Sadness Glossary

GRIMACING
Uncomfortable, nervous, worry

SURPRISED
Unanticipated, curious

STUNNED
Shocked, amazed, surprise

PENSIVE
Sad, disappointed, thoughtful

THINKING
Pondering, questioning, wonder

HAND OVER MOUTH
Shocked, embarrassed, surprise

YAWNING
Tired, bored, sleepy, exhausted

PEEKING
Shy, embarrassed, bashful

RAISED EYEBROW
Sceptical, suspicious, doubtful

UNAMUSED
Unimpressed, disapproving

EYES ROLLING UP
Exasperation, disbelief, annoyed

EXHALING
Frustrated, exasperated, tired

WIDE EYED
Sad, pitiful, desperate

WORRIED
Anxious, concerned, troubled

CRYING
Disappointed, hurt, raw

UNHAPPY
Depressed, upset, solemn

FROWN
Sad, dissatisfied, discontent

UNSURE
Awkward, uneasy, unexpected

CONFUSED
Puzzled, perplexed, uncertain

BLUSHING
Embarrassed, shy, bashful

MIND BLOWN
Amazed, blown away, incredible

DOWNCAST SWEAT
Stressed, worried, frustrated

DISAPPOINTED
Disheartened, forlorn, glum

PERSEVERING
Struggling, determined

DROOLING
Tempting, attractive, yummy

UPSIDE DOWN
Sarcasm, awkward, uneasy

SHUSHING
Quiet, secret, hush, confidential

ZIPPER MOUTH
Muted, silent, secretive

CONFUSED
Disorientated, dizzy, unsure

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Despair Glossary

ANXIOUS
Worried, stressed, panicked

FEARFUL
Scared, frightened, shocked

DISMAYED
Distressed, anguished, alarmed

ANGUISHED
Shocked, upset, distressed

SCREAMING
Frightened, terrified, shocked

WAIL
Exhausted, frustrated, tired

DESPAIR
Fatigued, irate, exhausted

ANGRY
Mad, cross, indignant, hostile

POUTING
Furious, very angry, outraged

CURSING
Rage, swearing, frustrated

FURY
Angry, frustrated, fuming

SLEEPY
Drowsy, tired, yawning, weary

DISENGAGED
Unresponsive, disoriented, lost

DISASSOCIATED
Empty, blank, numb, absent

☑ Final Checklist

- ☐ **Create comfortable environment**
  Free from judgement and confusion
- ☐ **Build trust through rapport**
  Use stories and relatable scenarios
- ☐ **Track progress and development**
  Encourage variety and experimentation
- ☐ **Offer the use of headphones**
  Help students focus on their own sounds
- ☐ **Record any issues / complications**
  Problems can be resolved if reported
- ☐ **Feedback form completed**
  Gather opinions & suggestions
- ☐ **Each guardian given take-aways**
  With links, info and further reading

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

FEEDBACK

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Initial Student Moods

How are you currently feeling?



How do you feel about new things?



How do you feel listening to music?



If you could be more musical, how would that make you feel?



How do you feel about playing with others?



How would you feel if you were in a pop group?



Review

How many stars out of 5 would you give this session ☆☆☆☆☆

Try and ascertain how the session was experienced and appreciated, both during the experience and reflectively at the end.

Consider the impact on :

- Educational aspect
- Clinical /well-being sense
- Growth, personal development
- User experience
- Continuation at home
- Their interests

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

During:

- How did they find the visuals and the music?
- Was the volume right, and experience stimulating and fun, or was it overwhelming?
- Did they enjoy the session, was it satisfying?

Reflectively :

- Did they benefit and learn?
- Did they consider it worthwhile?
- How was the engagement with the other practitioners and the social aspect?
- Did they develop any new skills?

Worksheet

Enjoyment

How much did they enjoy it? ☆☆☆☆☆

How aligned were the sessions to their interests? ☆☆☆☆☆

How enthusiastic were they? ☆☆☆☆☆

Confidence

How well did the child understand the activities? ☆☆☆☆☆

How successful was the student in completing the activities? ☆☆☆☆☆

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

How suitable were the activities for their abilities? ☆☆☆☆☆

Inclusion

How well did the session help the student feel included? ☆☆☆☆☆

How well did the students work together? ☆☆☆☆☆

How well did the student stay on task (focused attention)? ☆☆☆☆☆

How suitable was the session for the individual's needs? ☆☆☆☆☆

Engagement

How would you rate their engagement with the group? ☆☆☆☆☆

How positive was the student's overall response to the session? ☆☆☆☆☆

How useful was the session for the student's learning and development? ☆☆☆☆☆

Communication

How social was the session? ☆☆☆☆☆

How would you rate this session? ☆☆☆☆☆

How much did they benefit from the session? ☆☆☆☆☆

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Student Questions

Do you like music?	☐ YES	☐ NO
Did you make lots of faces?	☐ YES	☐ NO
Did you enjoy the activity?	☐ YES	☐ NO
Did you have fun?	☐ YES	☐ NO
Would you like to do this activity again?	☐ YES	☐ NO
Did you enjoy making music?	☐ YES	☐ NO

Student Feedback
Draw in the boxes

Draw your favourite sounding facial expresssion

Draw your current mood

How did you feel when your friend made a face?

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Other Feedback
Complete the worksheet

How would you describe your experience?

What did you find to be the most valuable part of the session?

What did you find out?

Is there anything you would like to change? What was fun / boring

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Student Feelings

What is your current mood?



What was your mood during the session?



How did the music make you feel?



How did playing with others make you feel?



How did playing with others make you feel?



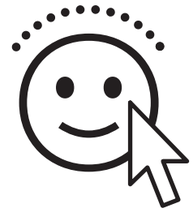
How would you feel about another session?



OPERATION

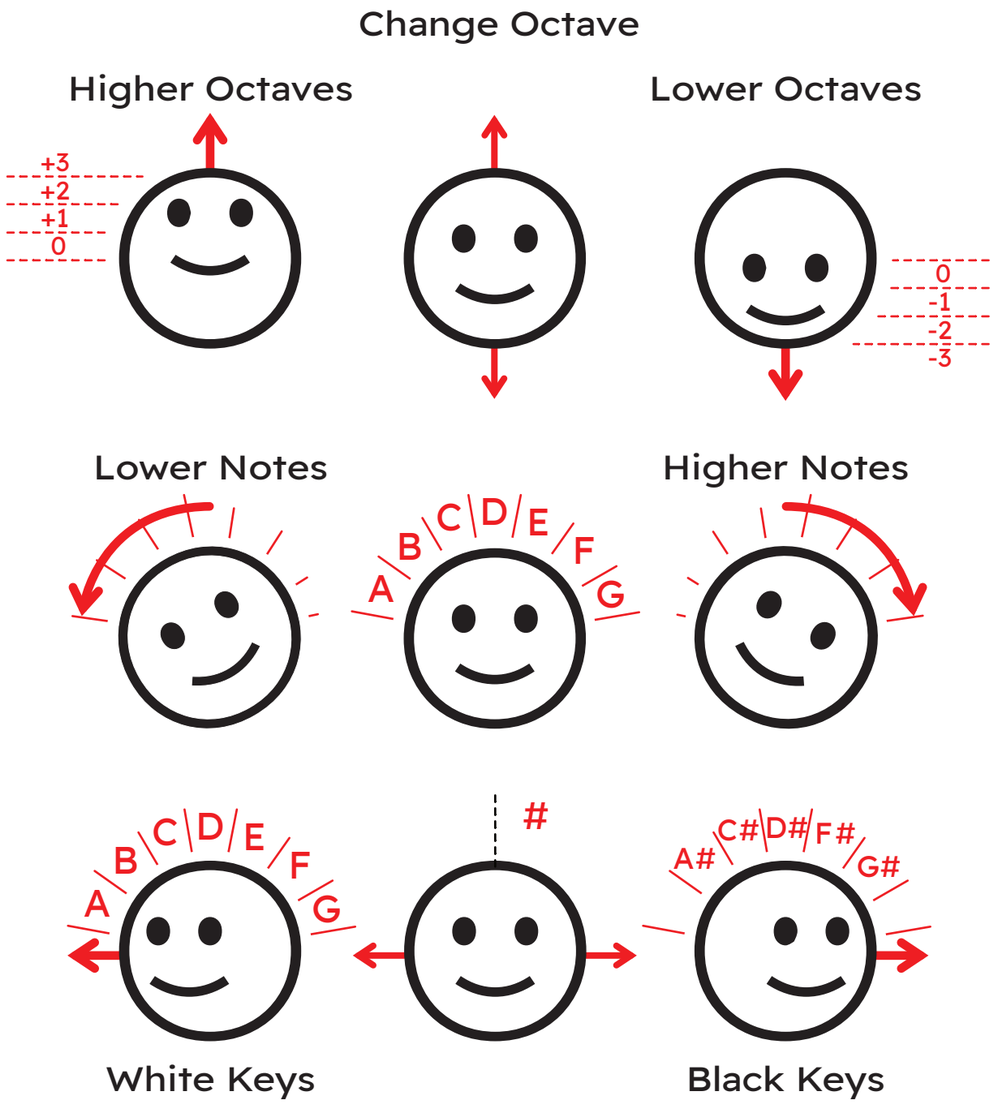
THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Play Modes
Distinct musical options



Change Play Mode

Chromatic Mode



THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Sympathetic Mode

A way to use facial expressions to guide the mood of the music - smile for a happy sound, frown for meloncholly.

Expressions are based on emojis which aid in the communication and simplify the operation. Some examples of common expressions and how they sound :



Smiling
Happy
Major Chord



Unhappy
Sad
Minor Chord



Grinning
Very Happy
Ionian Chord

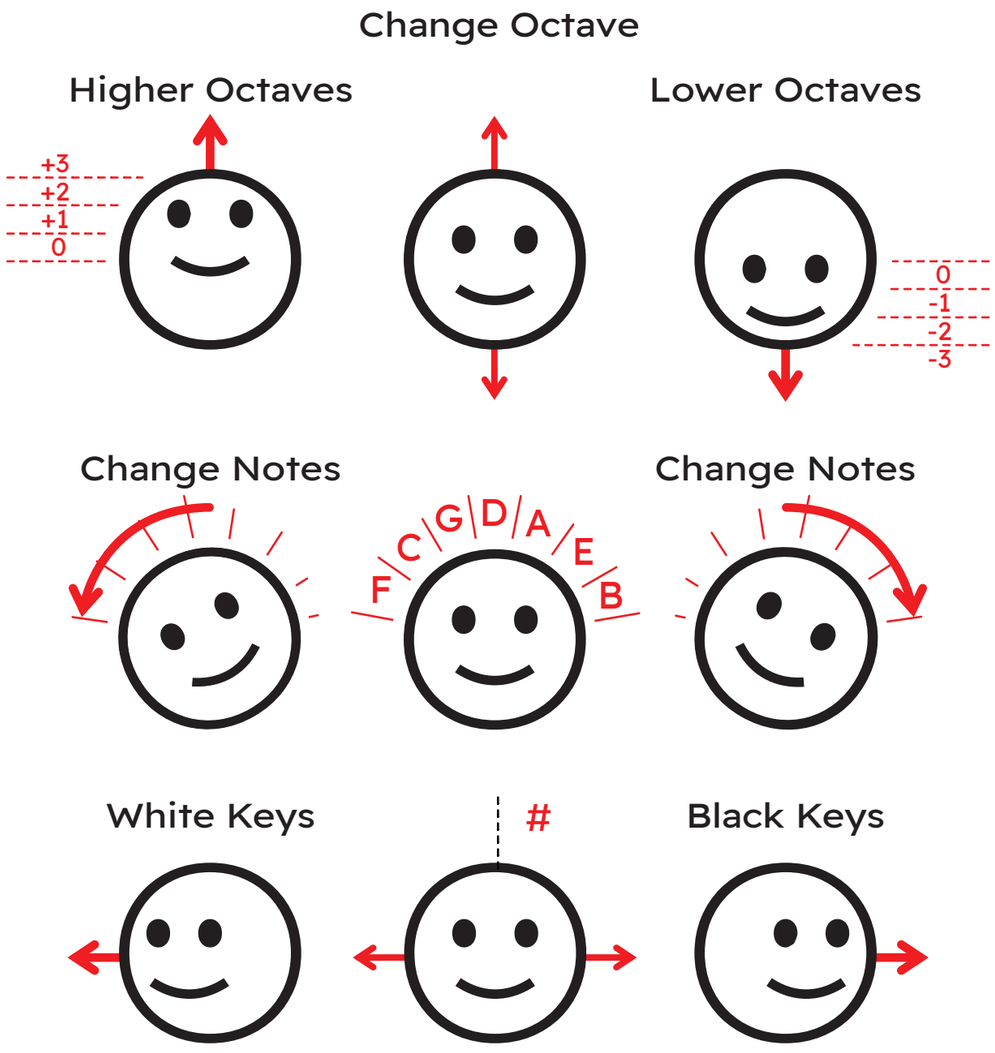


Frowning
Very Sad
Locrian Chord

Pan Audio
By moving your eyes



Circle of Fifths Mode



EQUIPMENT

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Setup

The session requires a safe space, uncluttered with enough room for people to be able to dance around.

Depending on the quantity of participants, a queue system can be considered, which if combined with activities can entertain big sessions by managing small dedicated groups.

Tech Requirements

- **Access to power sockets**
  Minimum 3 electrical Sockets (power strip will be provided so only 1 onsite required).
- **Amplified Audio & Hearing loop**
  Such as BT speakers or active loudspeakers. Or optionally comfortable headsets.
- **Good continuous lighting**
  For facial recognition and motion capture.
- **Computer machine with a fast GPU / NPU**
  Powerful CPU is ideal (eg. mac mini M4) Only 2Gb minimum RAM requirement.
- **An auto-focusing video camera / webcam**
  Preferably with a low light mode and fast refresh rate, high definition not required.
- **GPU powered web browser**
  Such as Google Chrome / Brave / Edge.
- **Mouse / track-pad + keyboard**
  For initial setup & active controls.
- **Screen or projector**
  See next page for specific setup diagrams
- **Internet connection required**
  To download software only.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Setup Considerations

The software takes the video feed from the camera, flips it to make the mirror image, then shows that on the screen, so it is important to position the camera close to the screen, ideally at eye level.

With bigger screens the camera can be placed directly in front of the screen.

Sound is delivered in stereo where available.

Connection to other electronic musical devices is possible via MIDI (see dedicated MIDI equipment section).

Screen / Projector

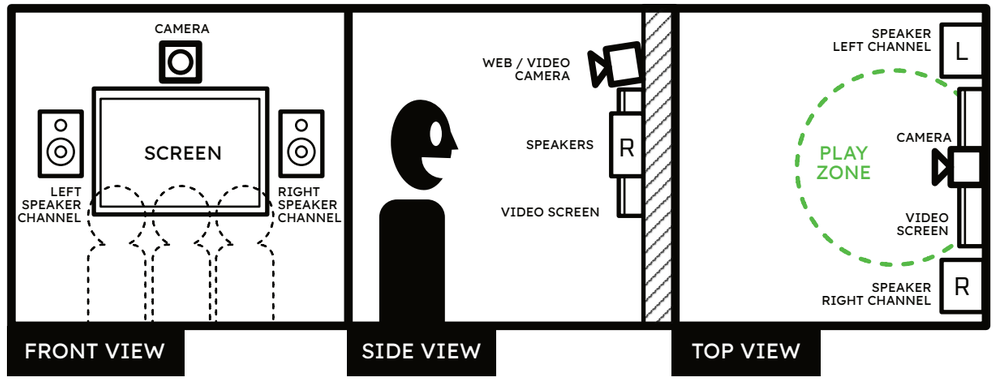
The primary interface engages people like a mirror does, with the idea to make the effect convincing so that everybody sees themselves and immediately is immersed.

Whilst a screen is ideal, it is also possible to use projectors but they must be positioned above the participants for long range models, and inbetween the participants and the screen for short throw models. The larger the screen, the more immersive and captivating the experience is, even greater than life-size works very well.

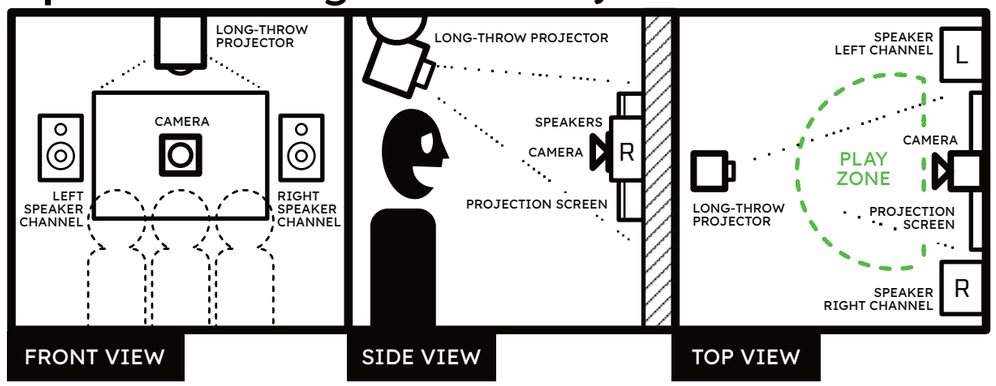
A HDMI Cable is required to connect the computer to the screen. If your machine does not have a HDMI-Out port, a USB dongle with HDMI will be required.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

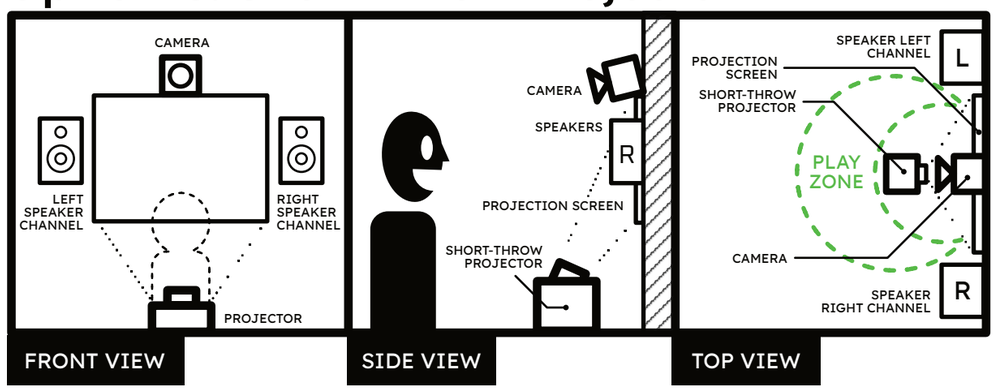
Option A. Video Screen / Monitor



Option B. Long Throw Projector



Option C. Short Throw Projector



Sound

Stereo speakers and headsets are preferred but not required.

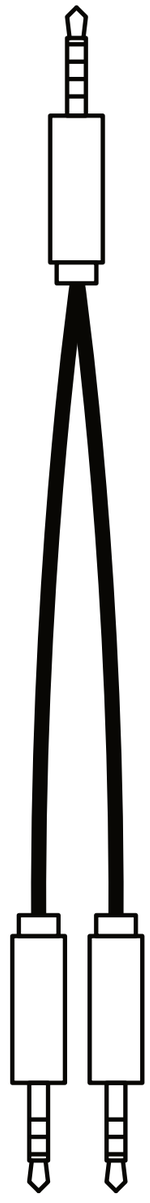
Quiet sessions can be created for multiple players using a 4 way headphone splitter *.

Bluetooth speakers are an option but consider the setup time and pairing the devices ahead of the session.

Bluetooth reduces cables but introduces a latency, a short time delay, that depending on your device may be noticeable.

Tip : Some eye actions affect the stereo panning (if available) so if you are using stereo speakers ensure they are the right way around.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS



A stereo splitting cable can be used, plugged into the stereo output jack (usually located on the computer), that splits the signal into left and right channel and allows for a pair of speakers to be used that respects the stereo separation.

Electronic Equipment Using MIDI Standards

Musical Instrument Digital Interface is a globally recognised standard for how to communicate musical data to control musical instruments remotely.

This is the universal musical standard protocol and features on most electronic musical instruments.

By connecting an instrument via it's MIDI port, it can be made to play any tune or melody without touching it. Even MIDI software can be controlled. Devices connect via a USB cable, or via a dedicated MIDI interface that requires a MIDI cable and a MIDI adapter.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

This is very useful for groups as all musical devices can be connected together and share tempo and scale, harmonising output and making music easier.

Any MIDI enabled instrument can be connected to the PhotoSYNTH and controlled automatically. If there are multiple MIDI controllers connected, each player controls one each. Devices can be connected via USB or via a MIDI Adapter.

You can learn more about MIDI : midi.org

Tip : MIDI is chainable and extendable so many musical devices can be connected to one port by connecting to their MIDI through ports.

Camera Choice

FRAME RATE

Cameras come in a range of frame-rates (the amount of photographs it takes in a second) which affects how quickly the screen is updated. Fast refresh rates look more natural and make the movement on screen more smooth. 30FPS is the most common, but cameras can be bought with 60 or even 120 frames per second (sometimes referred to as 120Hz).

ADAPTIVE / LOW LIGHT MODE

Low light modes, sometimes called adaptive light sensors, or represented by their larger CMOS values, are cameras that can see well in different light conditions which is very handy, especially for areas

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

where the conditions cannot be controlled.

AUTO-FOCUS

Auto-focus is a feature that allows the camera to adjust its focus depending on what it can see. This is very useful in dynamic groups and for rooms with difficult lighting. Focal length is important To get right, and auto-focus handles that automatically for you.

AUTO-FRAMING (PTZ)

Pan, tilt, zoom cameras on motorised gimbals, designed to locate humans and focus the frame onto them. In theory these should be a good complement to the setup for use cases involving wheel chairs or beds, but be careful, some are not as reliable as others and even premium models can act erratically when confused.

Camera Positioning

The camera is the primary interface and so it's position is significant to the operation. It should be positioned at the groups' average head height so that everybody is in view.

Positioning at eye level can be as simple as mounting the camera above the screen but benefits from being mounted on a extendable / retractable pole, that can be raised or lowered easily during the session.



THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Tripods can be used but a solid, weighted base is preferred due to stability and the speed and ease to raise and lower the camera (ideally with one hand). Access to the camera is critical so do not put it outside of reach.

With large screens, it is wise to put the camera in front of the screen, preferably on a weighted telescopic stand. These stands do not have to extend beyond 1.5 metres tall.

The trailing cable must always be safely tucked away and not present a trip hazard.

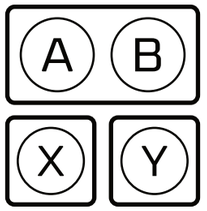
Tip : Always test the camera out before trying it in a group session.

Camera Recommendations

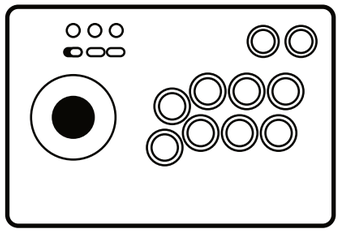
- **Osbot** Meet / Tiny 3
- **Razer** KIYO Pro / Ultra / X
- **Logitech** Brio 505 / 500 / 505e
- **Logitech** C920e Pro / StreamCam
- **Elgato** Facecam 4K / Facecam Neo
- **Elgato** Cam Link
- **Insta360** Link 2 / Link 2 Pro
- **YOLOLIV** YoloCam S7 / S3
- **Anker** PowerConf C300/C500 series
- **Opal** Tadpole
- **AVerMedia** PW513 / CAM 513
- **Sony** / **Canon** mirrorless via UVC

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

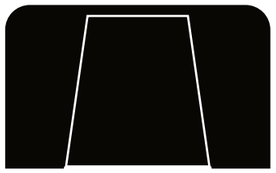
*Optional Accessories



Big tactile switches &
Logitech Adaptable Switches



Games Controllers,
Including Xbox Adaptive

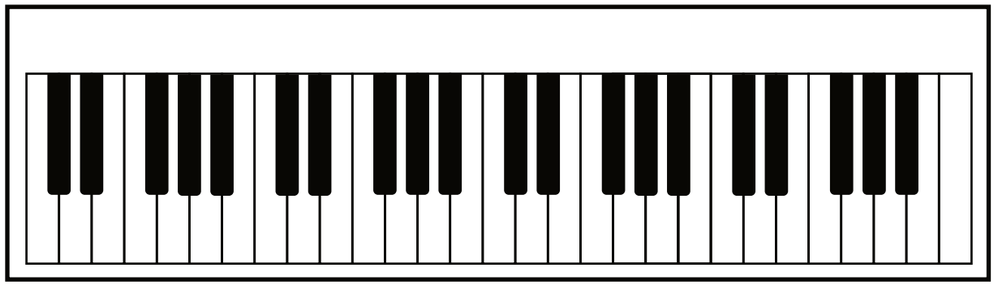


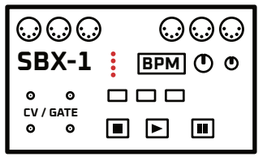
Floor Pedals



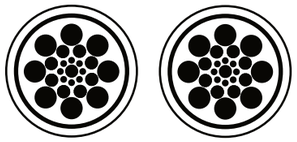
StreamDeck Button
Control Panel

MIDI Keyboard
NB. The ROLI Piano is fully supported

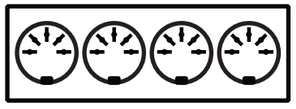




MIDI Adapter & Expander



Bluetooth (Stereo) Speakers



4 Way MIDI Splitter

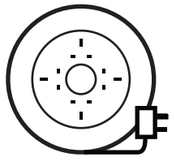
Tip : Cable ties, tunnels and cuffs can all be used to improve cabling.

Cables & Dongles

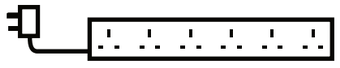
Sometimes you may find that you are missing an essential cable as they are very easy to lose. This is a list of useful spares.

- Power leads
- MIDI 1.0 Cables
- USB-C / USB-A / USB-C to USB-C
- USB Extender
- HDMI (mini/maxi/extender)
- Stereo-audio mini-jack to mini-jack
- Stereo Audio / Mic Extender
- USB Dongle with HDMI
- USB Hub for 8 devices
- Cable Sheathing

Useful Accessories



Power extension reel



6 Gang power strip



Hearing Loop



Lamps and ambient light



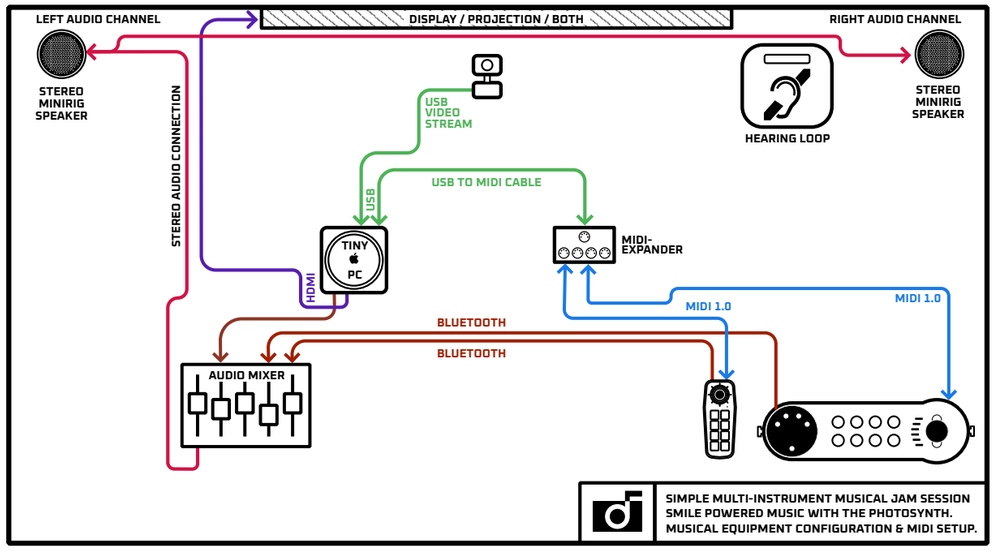
MAC mini M4 / NUC /
Laptop with GPU / NPU

Tip : Additional traditional musical instruments are great for groups!

Advanced Setups

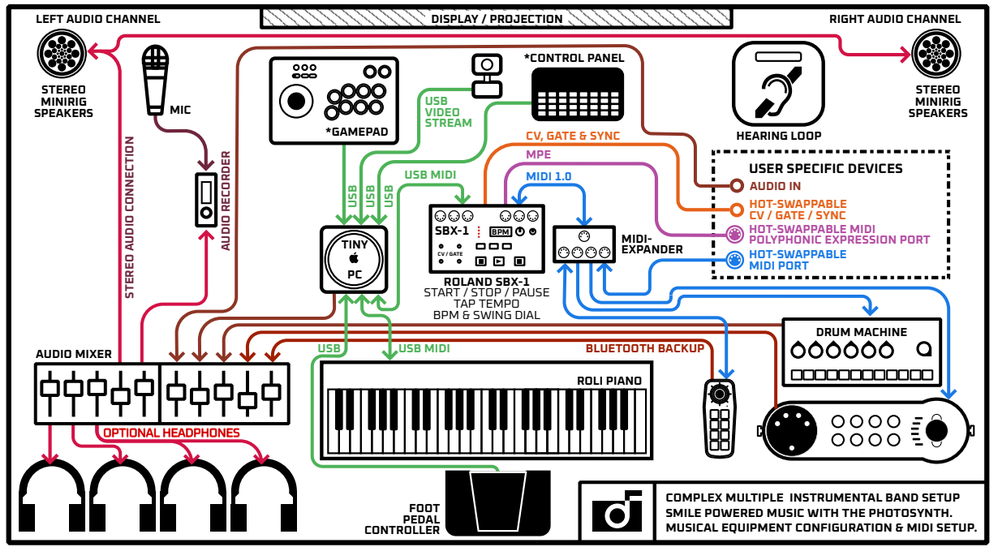
The software takes the video feed from the camera, flips it to make the mirror image, then shows that on the screen so it is important to position the camera as close to the screen as possible.

Multi-instrument Setup Example

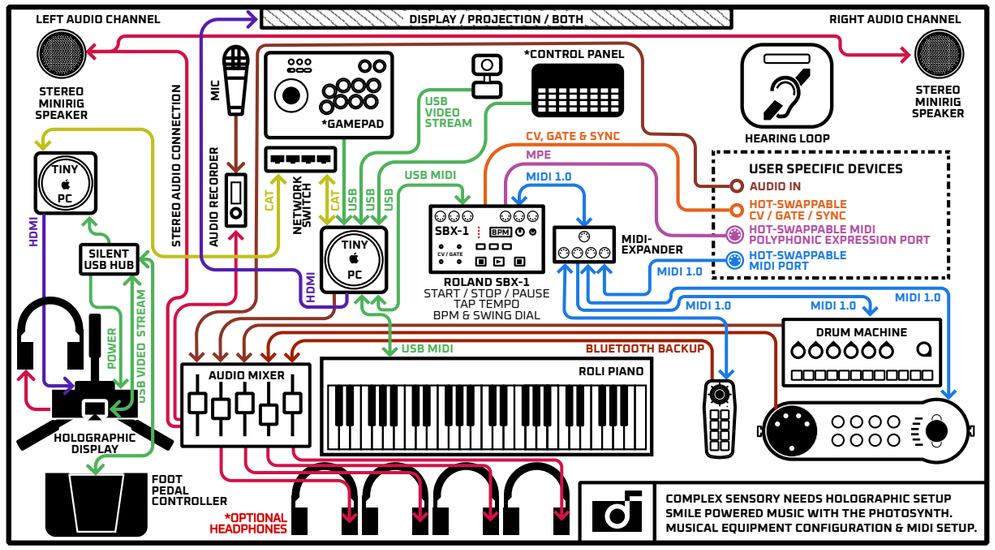


Example : 3 musical instruments playing together

Advanced Setup Examples



Example : Recorded session with other instruments



Example : Sensory Session with Holographic Display

Troubleshooting

On rare occasions, the software may act unexpectedly. Please read the information below for potential solutions.

Screen image is blurred and out of focus

Camera has likely lost focus and auto-focus is not recognising the people. Waving your hands and trying to catch the attention of the camera might kickstart the camera into focusing. If things do not improve, disconnect the USB camera and connect to another USB socket, restarting the browser and rerunning the software.

Multiple cameras are connected and the wrong one is being used

Click the settings button (bottom left in the control panel with a cog icon) and then select the camera from the list. If your camera is not listed, unplug and re-plug the camera into a different available USB socket. If it doesn't change on first select, try it again as sometimes cameras go to sleep if inactive.

Audio is crackly or has pops and clicks

It may be that your computer simply is not powerful enough to run the software unthrottled. Often laptops when they get too hot restrict the available resources in order to help them cool down, but this takes resources away from the software which results in the audio struggling to keep up with everything. Solutions may include cooling your computer with a heat-sink or fan, adding more memory and a stronger GPU and disabling all software running in the background.

People not being recognised

Lighting must be fixed and illuminating with a consistent glow. Sometimes strip lights cycle at frequencies similar to the refresh rates of monitors, if that is the case, the camera will appear to strobe. In these cases, it is best to turn off overhead CEF lights and instead use lighting with a constant illumination such as LED, incandescent or cold lamps.

No Sound being played

Ensure that the headphone socket is connected to the speaker and that the volume in the operating system is set high and not muted. There is a mute and volume section in the control panel of the software itself but it is best avoided in this instance.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Sound only coming out of one channel (left or right ear)

There is a stereo effect when you look left or right that is quite significant in its affect. If the person is angled to the camera, then their eyes may be panning the audio to one side. If they are far enough from the camera and central this should not affect them to that degree.

In single player mode, where multiple people are playing, the most prominent face is played - but this may not be the person wearing the headphones

The easiest fix would be to restrict the quantity of people playing or ensuring that the person with the headset is closest to the camera. We discovered interesting behaviours where kids were trying to "steal focus" which is a game I had not anticipated before. When this happens it is best to encourage them to experience the multiplayer version or to give them some takeaways such as the app to play at home.

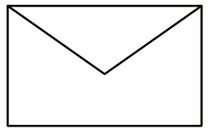
System has hung unexpectedly

Close the browser window and restart it. If you are still having issues, restart the machine.

Support

It is important to have support beyond the session, and to nurture any new skills, interests and discovered passions.

The following directory outlines some of the public groups that specialise in accessible music technology and support.



Contact about PhotoSYNTH
zenon@designerzen.com
+44 02031 297 441

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Support Directory

The following directory outlines some of the public groups that specialise in accessible music technology and support.

- Music Hubs
- Music Communities
- Music Workshops
- Social Enterprises
- Music Charities
- Specialist schools
- Accessible technology producers

Music Communities

Communities are groups of people with similar passions who come together to share these interests. There are many types of music communities, some are for very specific people, others are open to all.

Community spirit and support are beneficial for growth and development as well as the social side of meeting people with similar interests and interacting in wonderful ways.

Some communities meet regularly, especially local ones. There is usually a good mixture of ages and abilities.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Social Enterprises

Social Enterprises are businesses that prioritise social objectives through sustainable business practices, reinvesting profits back into their missions.

They often focus on community development and improving lives, blending the principles of entrepreneurship with social responsibility.

The following are trusted enterprises who have a proven track record with the public.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

GLOBAL MUSIC VISIONS C.I.C.

A user led social enterprise based in Portsmouth in the UK, facilitating inclusive music workshops for people of all ages and abilities, using music, technology, and local community spaces.

Services	Workshops
Contact	David Shervill
Phone	07742 311 486
Email	globalmusicvis@aol.com
Socials	@GlobalMusicVis
Website	www.globalmusicvisions.co.uk

ALLSTARS

Music tech tutoring across the north with a focus on electronic music production and accessibility.

Services	Workshops
Contact	Ross Elliott
Phone	
Email	ross@allstarents.co.uk
Socials	
Website	allstarents.co.uk

Music Hubs

Music Hubs are collaborative organisations that aim to enhance music learning opportunities and community connections, providing access to instruments, lessons, and performance experiences.

These hubs often work with schools, local authorities, and music providers to create a comprehensive music education system.

They are usually great sources for resources, connections and homes for music communities.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Music Charities

Charities are organisations set up to provide aid and support, funded mainly through donations, they work to improve lives and address systemic issues such as access to music technology.

DRAKE MUSIC

Providing support at the intersection of music and technology following the social model of disability.

Services	Support
Contact	Tim Yates
Phone	020 7739 5444
Email	info@drakemusic.org
Socials	@drake_music
Website	www.drakemusic.org

SOUND CONNECTIONS

Our mission is to ensure that all children and young people can experience music and creative opportunities that are relevant to them and responsive to their needs in a changing world.

Services	Youth Voice and Participation
Contact	Emily Perryman
Phone	020 7729 7220
Email	info@sound-connections.org.uk
Socials	@Sconnections
Website	www.sound-connections.org.uk

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

SOUND WITHOUT SIGHT

Community-led initiative, connecting blind and partially sighted musicians, audio engineers, and industry professionals to one another, and to resources, opportunities, and support.

Services	Access and research
Contact	Jay Pocknell
Address	Royal National Institute for the Blind
Email	contact@soundwithoutsight.org
Socials	@SoundWithoutSight
Website	soundwithoutsight.org

OMNIMUSIC

Our name means music for all, and that's exactly what we believe in. We deliver accessible music sessions to Disabled children and adults in partnership with local schools and community groups.

Services	Workshops and Sessions
Contact	Chris Ball
Address	23 High Street, Stockport, SK1 1EG
Email	info@omnimusic.org.uk
Socials	@omnimusicuk
Website	omnimusic.org.uk

OHMI - ONE HANDED MUSICAL INSTRUMENT

Pioneering the development and adaptation of musical instruments for people who are physically disabled.

Services	Research & Development
Contact	Andrew MacPherson
Address	The OHMI Trust, 29 Woodbourne Road, Harborne, Birmingham B17 8BY
Email	admin@ohmi.org.uk
Socials	@TheOHMITrust2016
Website	www.ohmi.org.uk

Specialist Schools

Educational institutions that focus on teaching subjects by offering tailored programs with unique support.

There are schools that specialise with different members of the community, encompassing the wide variety of human experiences and covering the broad diversity in abilities. They often have partnerships with relevant industries, providing students with practical experiences and career pathways.

HIGH PEAK SCHOOL

Community-led initiative, connecting blind and partially sighted musicians, audio engineers, and industry professionals to one another, and to resources, opportunities, and support.

Phone	0203 823 3030
Address	Kedleston Group Limited, Unit 7 Brook Business Centre, Cowley Mill Road, Uxbridge, UB8 2FX
Email	info@kedlestongroup.com
Socials	@kedlestongroup
Website	www.kedlestongroup.com

JOHN F KENNEDY SPECIAL SCHOOL

A special school with dedicated sites for profound and multiple learning difficulties (PMLD) and autism spectrum conditions (ASC), full of fantastic, vibrant and incredibly talented students.

Phone	020 8534 8544
Address	JFK Special School Pitchford Street, London, E15 4RZ
Email	info@jfk.lihtrust.uk
Website	www.jfkspecialschool.lihtrust.uk

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

Music Workshops

Workshops are interactive sessions aimed to facilitate participation in music, teaching skills and offering a safe space with a welcoming atmosphere and interesting people.

They typically involve hands-on activities where participants can learn from facilitators and collaborate with peers. Group activities can include jamming or singing.

Often there is an educational aspect to these workshops along with take home resources and exercises to practise.

Copyright

This work is copyrighted but licensed under a highly permissive, open-sourced agreement, AGL3.0 (https://www.gnu.org/licenses/agpl-3.0.en.html).

You are welcome to use and remix the code with the condition that all improvements must be shared with the public.

The source code is publicly available for free from github (https://github.com/designerzen/InterFACE/).

All rights reserved.
2011-2026 © Zenon Olenski / designerzen.

THE ROYAL SOCIETY SUMMER SCIENCE
EXHIBITS ON TOUR / PHOTOSYNTH & FRIENDS

References

If you would like to learn more about music access and inclusion, please check out some of these amazing projects and resources :

- AccessAble
  https://accessable.co.uk
- Attitude is Everything
  https://attitudeiseverything.org.uk
- The King's Trust
  https://www.kingstrust.org.uk/
- Scope
  https://www.scope.org.uk
- MenCap
  https://www.mencap.org.uk/
- Sense
  https://www.sense.org.uk/

PhotoSYNTH
Smile Powered Music

1. Try the software
Works best with Chrome or Edge browser on a modern device.
https://photosynth.app

2. Learn about the project
Science, technology and music theory is explained in detail.
https://photosynth.lol

Support

Email : zenon@designerzen.com
Call : +44 02031 297 441 (9AM-7PM)
Text (SMS) : +44 07877 239 954